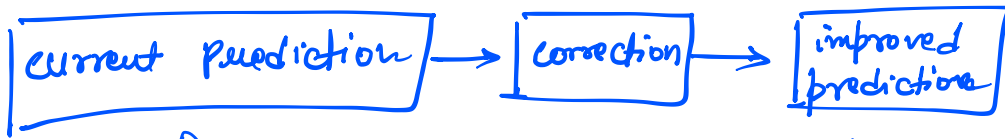


Boosting.

one extremely complicated model

or

improve a single model by fixing what it gets wrong.



Boosting:

$$F_m(x) = F_{m-1}(x) + \eta h_m(x)$$

$F(x)$ = f is the model, x is the input. model as a weak learner

m = iteration no.

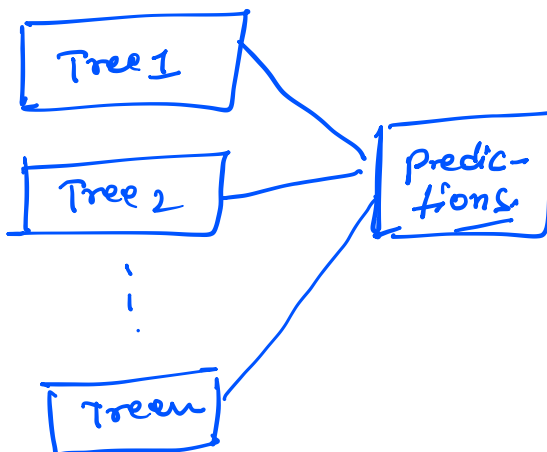
$h(x)$ = corrections on input x .

η $\rightarrow$ learning rate.

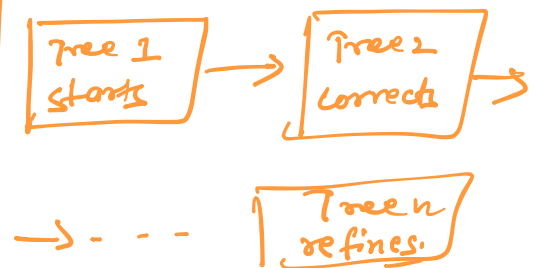
+ Bagging vs. Boosting:

Bagging:

$$\hat{F}_{\text{bag}}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$$



Boosting:



Boosting!

$$F_m(x) = f_0(x) + \eta h_1(x) + \eta h_2(x) + \dots + \eta h_m(x)$$

$$0 < \eta \leq 1$$

Deliveries	Distance x	Actual time y
A	1 km	2 min
B	2 km	8 min
C	3 km	10 min
D	4 km	12 min

① start with a weak learner $\rightarrow$ mean prediction

$$F_0(x) = \bar{y} = 8$$

compute residuals,

$$r_i^{(1)} = y_i - f_0(x_i)$$

Residuals

-6	0	2	4
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$$\begin{aligned} \text{MSE}(F_0) &= \frac{1}{4} (-6^2 + 0^2 + 2^2 + 4^2) \\ &= 14 \end{aligned}$$

② Correction:

new tree being trained on residuals.

$(1, -6), (2, 0), (3, 2), (4, 4)$

candidate thresholds: 1.5, 2.5, 3.5

Each threshold will split data into two parts

Decision Tree Split:
$$SSE_{\text{split}} = \sum_{i \in L} (r_i - \bar{r}_L)^2 + \sum_{i \in R} (r_i - \bar{r}_R)^2$$

L = left split

R = right split

$\bar{r}_L$ = mean of residuals of left split

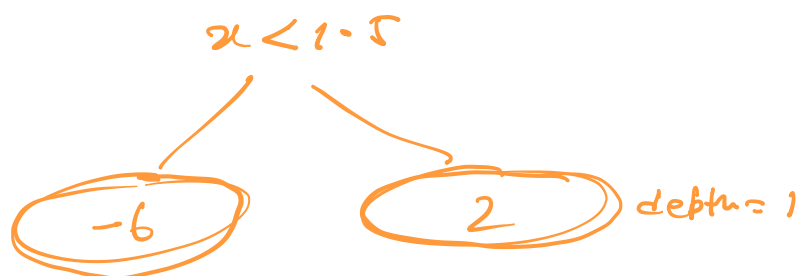
r_i = residual at i .

threshold 1.5:

$$\bar{r}_R = \frac{0 + 2 + 4}{3} = 2$$

$$SSE = \underbrace{(-6 - (-6))^2}_{\text{Left split}} + \underbrace{(0-2)^2 + (2-2)^2 + (4-2)^2}_{\text{Right split}}$$

$$= 8$$



choosing $\eta = 0.5$

$$f_1(x) = f_0(x) + 0.5 h_1(x)$$

$f_0(x)$	$f_1(x)$	<u>new residuals</u>
8	$8 + 0.5(-6) = 5$	-3
8	$8 + 0.5(2) = 9$	-1
8	9	+1
8	9	+3

$$\text{MSE}(f_1) = \frac{(-3)^2 + (-1)^2 + (1)^2 + (3)^2}{4}$$
$$= 5$$

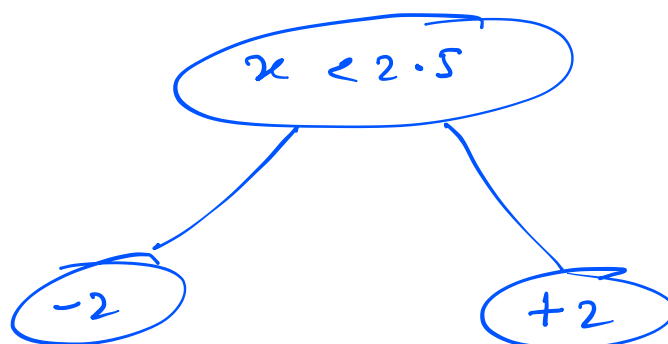
for f_2 .

$$(1, -3) (2, -1) (3, 1) (4, 3)$$

Train a decision tree on this sample.

$$\text{cand. thresh} = 1.5, 2.5, 3.5$$

$t = 2.5$ gives min. SSE.



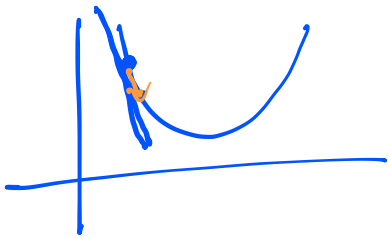
$$\eta = 0.5$$

$f_1(x)$	$f_2(x) = f_1(x) + 0.5 h_2(x)$	y
5	$5 + 0.5(-2) = 4$	-2
9	8	0
9	10	0
9	10	2

$$MSE(f_2) = \frac{(-2)^2 + 0^2 + 0^2 + (-2)^2}{4}$$

$$= 2$$

$$f_2(x) = \underbrace{f_0(x)}_{\text{mean}} + 0.5 \underbrace{h_1(x)}_{\text{Trained on res.}} + 0.5 \underbrace{h_2(x)}_{\text{Trained on res.}}$$



the derivative :- increasing the prediction, increases the loss,

-ve derivative: increasing the prediction, decreases the loss,

$$\gamma_{im} = \frac{\partial L(y_i, F(x_i))}{\partial F(x_i)} \bigg|_{F = F_{m-1}}$$

$$L(y, F) = \frac{1}{2} (y - F)^2$$

$$\frac{\partial L}{\partial f} = \frac{1}{z} \cdot z \cdot (y - f) \cdot -1$$

$$= -(y - f)$$

$$\boxed{-\frac{\partial L}{\partial f} = y - f} \rightarrow \text{residuals.}$$

losses	gradient	significance
MSE	$y - f$	Large errors strongly penalized
MAE	kind of a sign function.	Penalizes outliers.

* classification:

$$f_0(x) = \text{prior}$$

$$f(x) = f_0(x) + \eta h_1(x) + \eta h_2(x) + \dots + \eta h_n(x)$$

$$p(x) = \sigma(f(x))$$

$\hookrightarrow$ sigmoid.

$$\sigma(x) = \frac{1}{1 + e^{-x}} \in (0, 1)$$

Apply threshold.

$$L(y, P) = -y \log(P) - (1-y) \log(1-P)$$

compute gradient. Update model.